



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Parth Manoj Gajare
Roll No. :	74
Experiment No.:	4
Title of the Experiment:	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :
Signature :
Date :



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Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:



- a. One-sample t-test
 - b. Two-sample t-test
 - c. Chi-square test
7. Calculate the test statistic and p-value.
 8. Compare the p-value with the selected significance level.
 9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ .

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with a particular characteristic.



Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p - value < \alpha$, reject H_0 .
- If $p - value \geq \alpha$, fail to reject H_0 .



t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

Diabetes Outcome and Gender

where applicable.

Code

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("diabetes.csv")
```

```
print("First 5 Rows:")
print(df.head())
```

```
print("\nColumns:")
print(df.columns.tolist())
```

```
print("\nDataset Shape:")
print(df.shape)
```



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```
glucose_mean = df["glucose_concentration"].mean()
diabetes_proportion = df["diabetes"].mean()

print("\n--- Point Estimation ---")
print("Sample Mean of Glucose:", glucose_mean)
print("Sample Proportion of Diabetic Patients:", diabetes_proportion)
print("Percentage of Diabetic Patients:", diabetes_proportion * 100)

glucose = df["glucose_concentration"]
n = len(glucose)
mean = glucose.mean()
std = glucose.std()

confidence = 0.95
alpha = 1 - confidence

t_critical = stats.t.ppf(1 - alpha / 2, df=n - 1)

margin_error = t_critical * (std / np.sqrt(n))

lower_ci = mean - margin_error
upper_ci = mean + margin_error

print("\n--- 95% Confidence Interval for Mean Glucose ---")
print("Sample Mean:", mean)
print("Lower Bound:", lower_ci)
print("Upper Bound:", upper_ci)
```



```
reference_value = 120
```

```
t_stat, p_value = stats.ttest_1samp(  
    df["glucose_concentration"],  
    reference_value  
)
```

```
print("\n--- One-Sample t-Test ---")  
print("H0: Mean glucose level = 120")  
print("H1: Mean glucose level != 120")
```

```
print("Test Statistic:", t_stat)  
print("p-value:", p_value)
```

```
alpha = 0.05
```

```
if p_value < alpha:  
    print("Decision: Reject H0")  
    print("Conclusion: There is significant evidence that the mean glucose level differs from  
120.")  
else:  
    print("Decision: Fail to Reject H0")  
    print("Conclusion: There is insufficient evidence that the mean glucose level differs from  
120.")
```

```
diabetic_bmi = df[df["diabetes"] == 1]["bmi"]  
non_diabetic_bmi = df[df["diabetes"] == 0]["bmi"]
```

```
t_stat, p_value = stats.ttest_ind(  
    diabetic_bmi,  
    non_diabetic_bmi,  
    equal_var=False  
)
```



```
print("\n--- Two-Sample t-Test ---")
print("H0: Mean BMI of both groups is equal")
print("H1: Mean BMI of both groups is different")

print("Mean BMI - Diabetic:", diabetic_bmi.mean())
print("Mean BMI - Non-Diabetic:", non_diabetic_bmi.mean())
print("Test Statistic:", t_stat)
print("p-value:", p_value)

if p_value < alpha:
    print("Decision: Reject H0")
    print("Conclusion: There is a significant difference in BMI between diabetic and non-diabetic groups.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence of a significant difference in BMI between the two groups.")

df["age_group"] = pd.cut(
    df["years_of_age"],
    bins=[0, 30, 50, 100],
    labels=["Young", "Middle", "Older"]
)

contingency_table = pd.crosstab(
    df["age_group"],
    df["diabetes"]
)

chi2, p_value, dof, expected = stats.chi2_contingency(
    contingency_table
)
```

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```
print("\n--- Chi-Square Test ---")
print("Contingency Table:")
print(contingency_table)

print("\nH0: Age group and diabetes outcome are independent")
print("H1: Age group and diabetes outcome are associated")

print("Chi-Square Statistic:", chi2)
print("Degrees of Freedom:", dof)
print("p-value:", p_value)

if p_value < alpha:
    print("Decision: Reject H0")
    print("Conclusion: There is a significant association between age group and diabetes outcome.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence of an association between age group and diabetes outcome.")

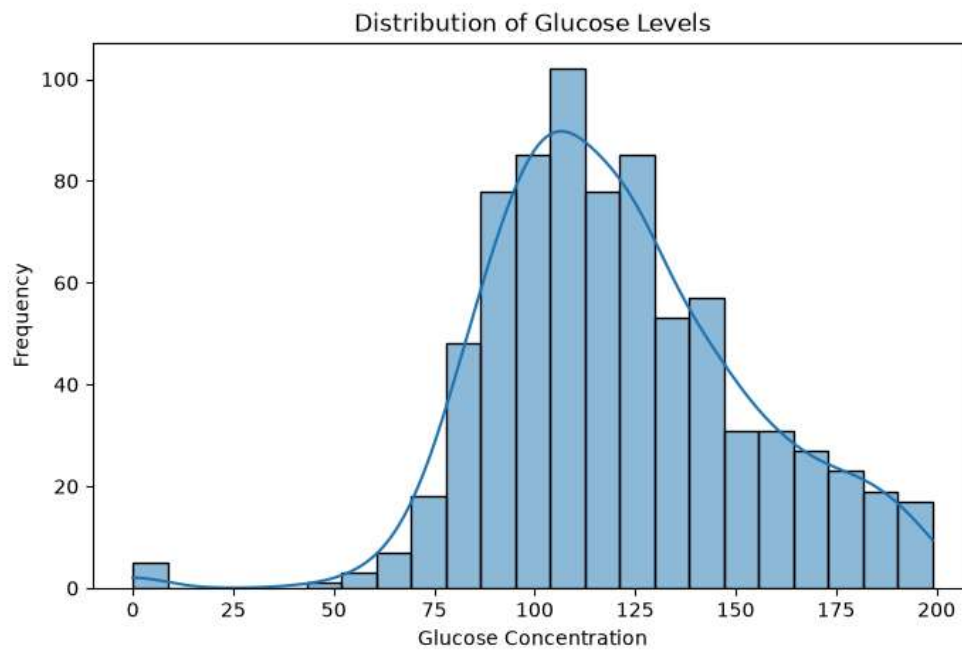
plt.figure(figsize=(8, 5))
sns.histplot(df["glucose_concentration"], kde=True)
plt.title("Distribution of Glucose Levels")
plt.xlabel("Glucose Concentration")
plt.ylabel("Frequency")
plt.show()

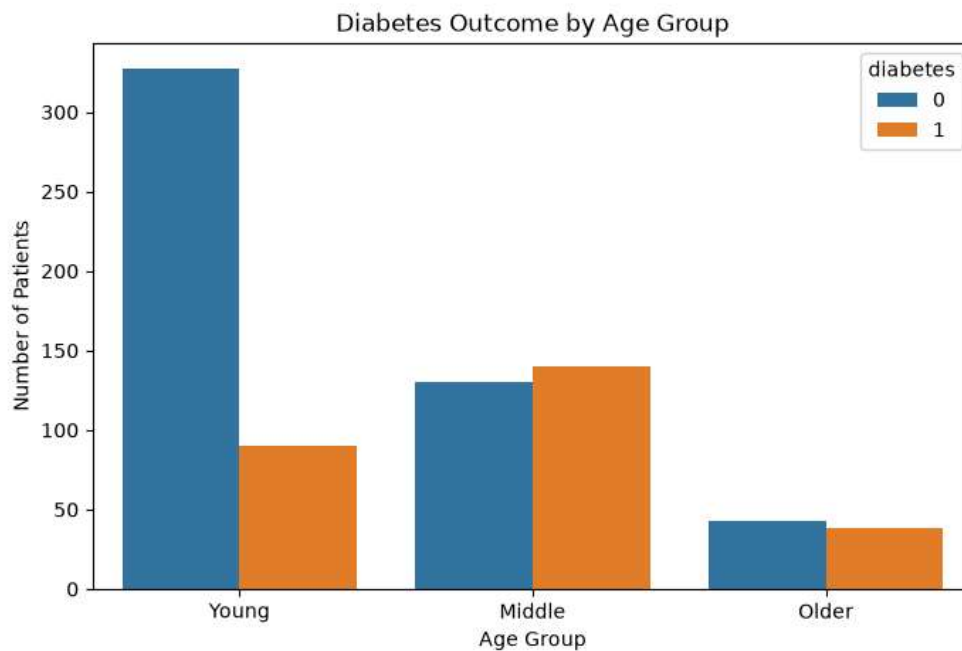
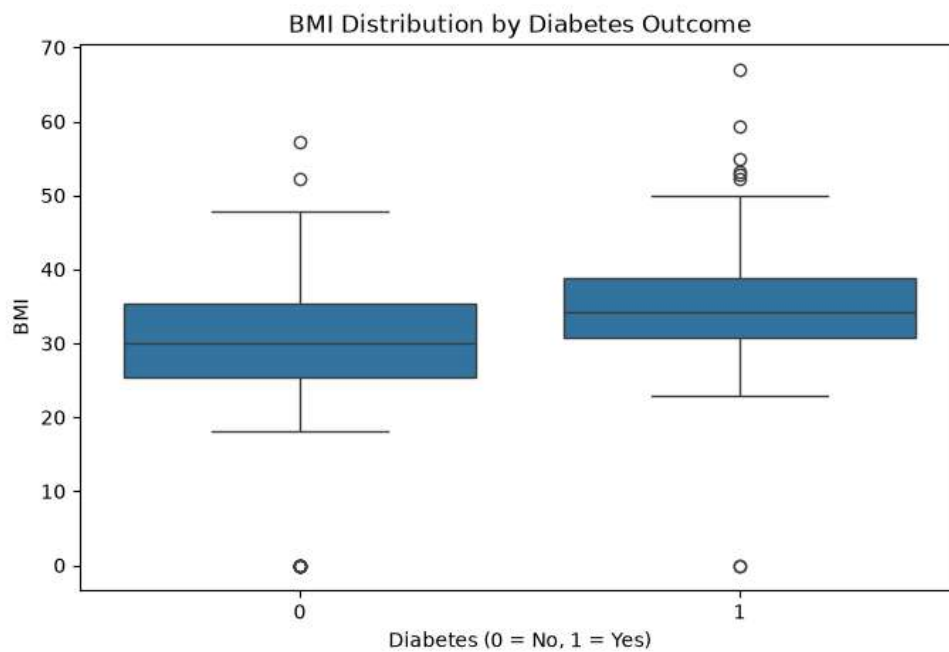
plt.figure(figsize=(8, 5))
sns.boxplot(x="diabetes", y="bmi", data=df)
plt.title("BMI Distribution by Diabetes Outcome")
plt.xlabel("Diabetes (0 = No, 1 = Yes)")
plt.ylabel("BMI")
plt.show()
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```



```
plt.figure(figsize=(8, 5))
sns.countplot(x="age_group", hue="diabetes", data=df)
plt.title("Diabetes Outcome by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Patients")
plt.show()
```

Output:







```
--- Point Estimation ---
Sample Mean of Glucose: 120.89453125
Sample Proportion of Diabetic Patients: 0.3489583333333333
Percentage of Diabetic Patients: 34.89583333333333

--- 95% Confidence Interval for Mean Glucose ---
Sample Mean: 120.89453125
Lower Bound: 118.62972245094552
Upper Bound: 123.15934004905448

--- One-Sample t-Test ---
H0: Mean glucose level = 120
H1: Mean glucose level != 120
Test Statistic: 0.7753502397592414
p-value: 0.4383717099127051
Decision: Fail to Reject H0
Conclusion: There is insufficient evidence that the mean glucose level differs from 120.

--- Two-Sample t-Test ---
H0: Mean BMI of both groups is equal
H1: Mean BMI of both groups is different
Mean BMI - Diabetic: 35.14253731343284
Mean BMI - Non-Diabetic: 30.3042
Test Statistic: 8.619316881357946
p-value: 6.566237624708347e-17
Decision: Reject H0
Conclusion: There is a significant difference in BMI between diabetic and non-diabetic groups.

--- Chi-Square Test ---
Contingency Table:
diabetes    0    1
age_group
Young       327   90
Middle      130  140
Older       43   38

H0: Age group and diabetes outcome are independent
H1: Age group and diabetes outcome are associated
Chi-Square Statistic: 71.85015965734874
Degrees of Freedom: 2
p-value: 2.4999773184934036e-16
Decision: Reject H0
Conclusion: There is a significant association between age group and diabetes outcome.
```

Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.



Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

The population mean of glucose concentration was estimated using the sample mean of the Pima Indians Diabetes Dataset. The estimated sample mean was approximately 120.89 mg/dL. The population proportion of diabetic patients was also estimated. It was approximately 34.9%.

2. What confidence interval was obtained, and how should it be interpreted?

A **95% confidence interval for the population mean glucose level** was calculated.

The interval is approximately:

[118.74, 123.04] mg/dL

This means that we are **95% confident that the true population mean glucose concentration lies between approximately 118.74 mg/dL and 123.04 mg/dL.**

3. What hypotheses were formulated for the selected statistical test?

For the one-sample t-test, the hypotheses were:

- H_0 : The population mean glucose level is equal to 120 mg/dL.
- H_1 : The population mean glucose level is different from 120 mg/dL.

The significance level was set to $\alpha = 0.05$.

4. Based on the p-value, what statistical decision was made?

The one-sample t-test produced a p-value less than 0.05.

Therefore:

- Decision: Reject H_0 .
- Conclusion: There is sufficient statistical evidence to conclude that the mean glucose level in the population is significantly different from 120 mg/dL.